

Training Configuration and Summary:

Dataset used: rssi_odom_dataset_12apr25_ajust.parquet

model_type: GRU
group_sec: 80
timesteps_orig: 60
batch_size: 256
min_units: 16
max_units: 1024
units_step: 16
dropout_min: 0.2
dropout_max: 0.3
dropout_step: 0.1
activation_functions: ['tanh', 'relu', 'elu']
optimizers: ['adam', 'rmsprop', 'Nadam']
min_layers: 1
max_layers: 2
max_epochs: 1000
hyperband_iterations: 3
group_msec: 80
samples_per_sec: 12
forecast_horizon_sec: 4
forecast_steps: 48
timesteps: 60

GRU Multi-Step Training Report

Report Generated: 2025-05-22 09:47:16

Training Start: 2025-05-22_06-48-11

Training End: 2025-05-22_09-37-20

Training Duration: 169.15 minutes

Predictions Start: 2025-05-22_09-37-20

Predictions End: 2025-05-22_09-39-14

Predictions Duration: 1.90 minutes

Best Hyperparameters:

Number of Layers: 2

Activation: tanh

Optimizer: Nadam

Dropout Rate: 0.20

MAE: 2.5588

MSE: 10.8074

RMSE: 3.2875

R2 Score: 0.3501

Multi-Step Forecast:

Forecast Steps: 48

MAE: 0.0354

MSE: 0.0020

RMSE: 0.0452

R2 Score: 0.7793

Dataset size before down-sample: 85,269 rows
Dataset size after down-sample: 12,829 rows

First 5 timestamps BEFORE:

1744397085760
1744397085770
1744397085789
1744397085801
1744397085817

...

Last 5 timestamps BEFORE:

1744464971366
1744464971377
1744464971390
1744464971402
1744464971414

First 5 timestamps AFTER:

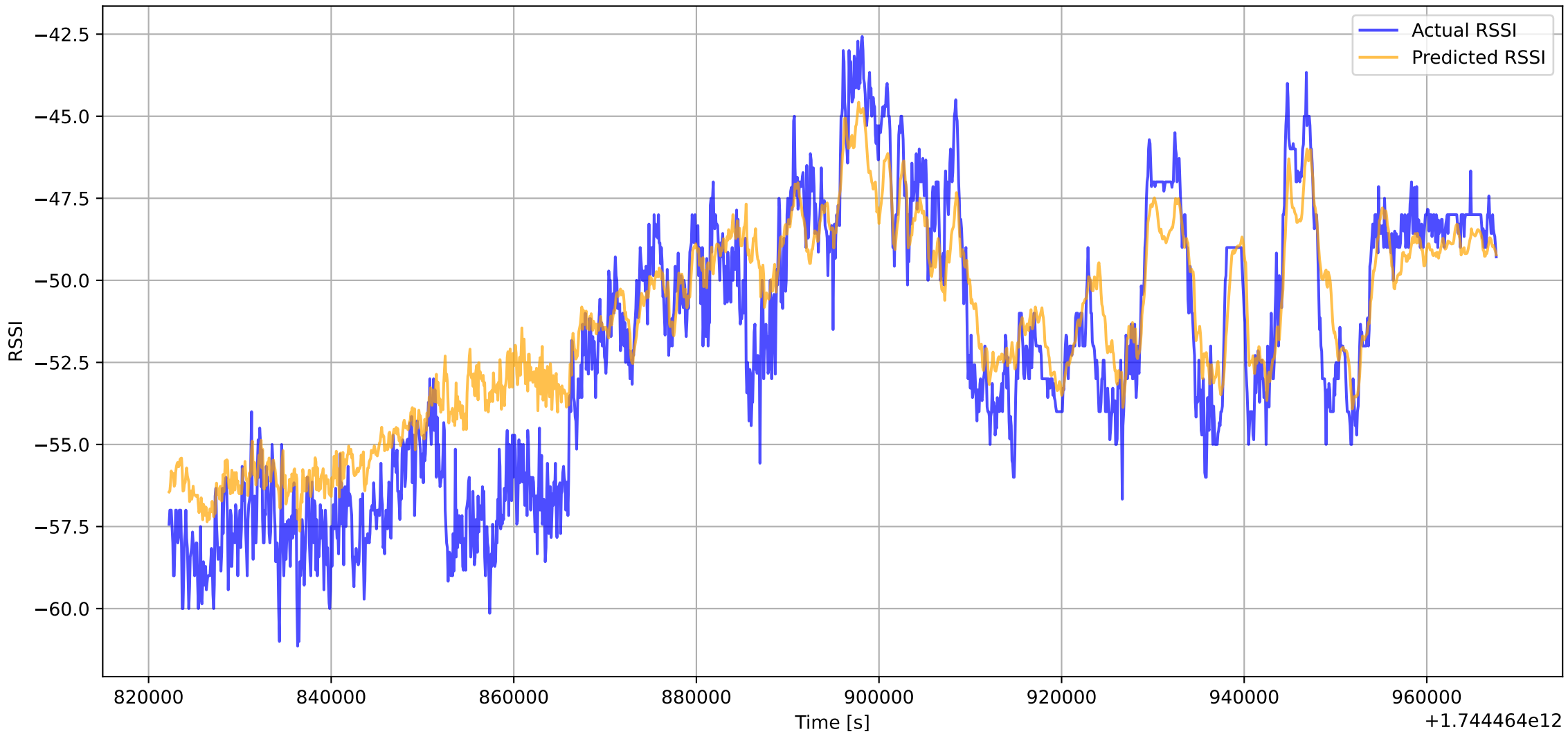
1744397085760
1744397085841
1744397086014
1744397086103
1744397086172

...

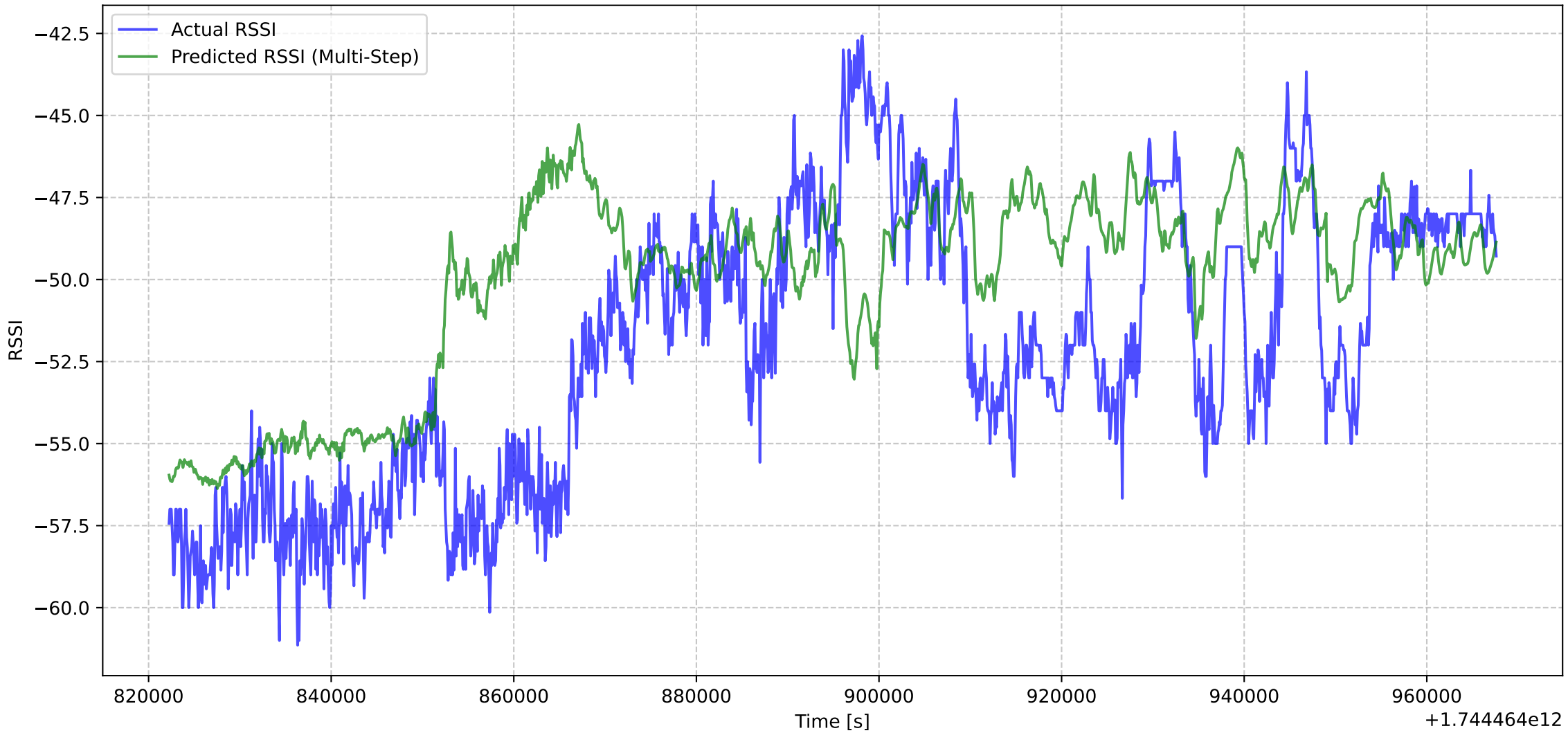
Last 5 timestamps AFTER:

1744464971041
1744464971124
1744464971209
1744464971281
1744464971366

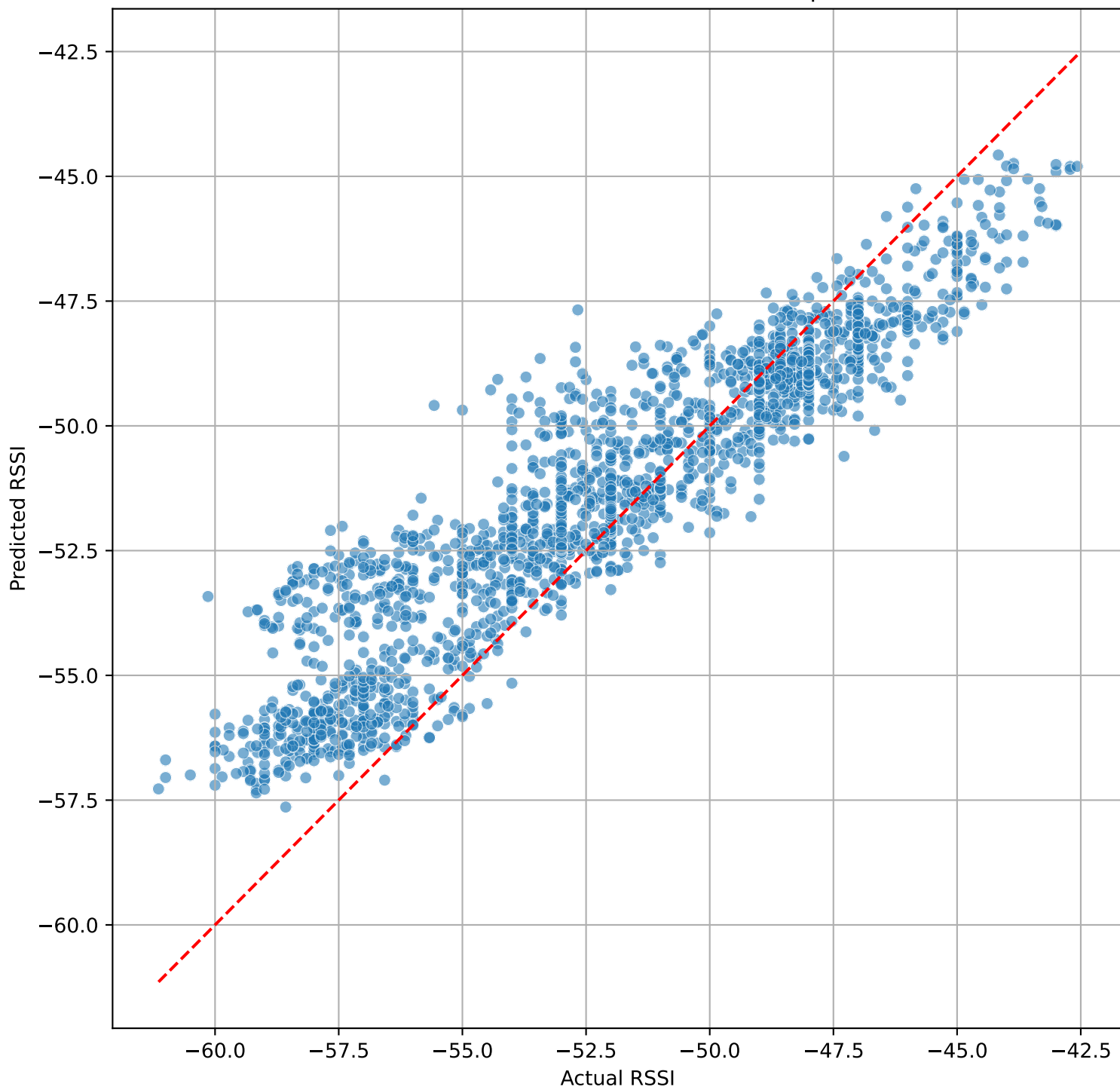
Actual vs Predicted RSSI (First Step)



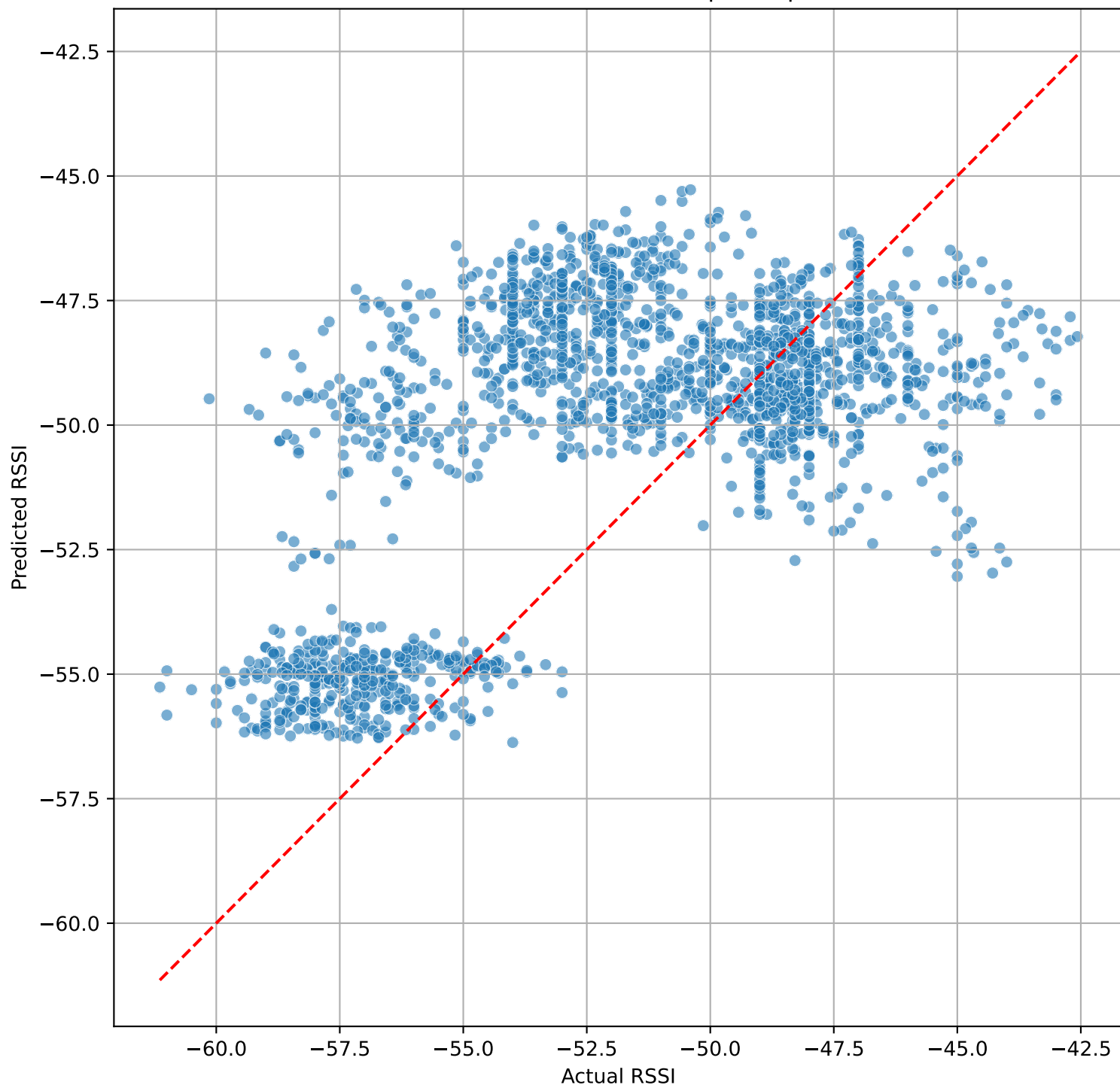
Actual vs Predicted RSSI (Multi-Step (step 48 $\approx$ 4.0 s))



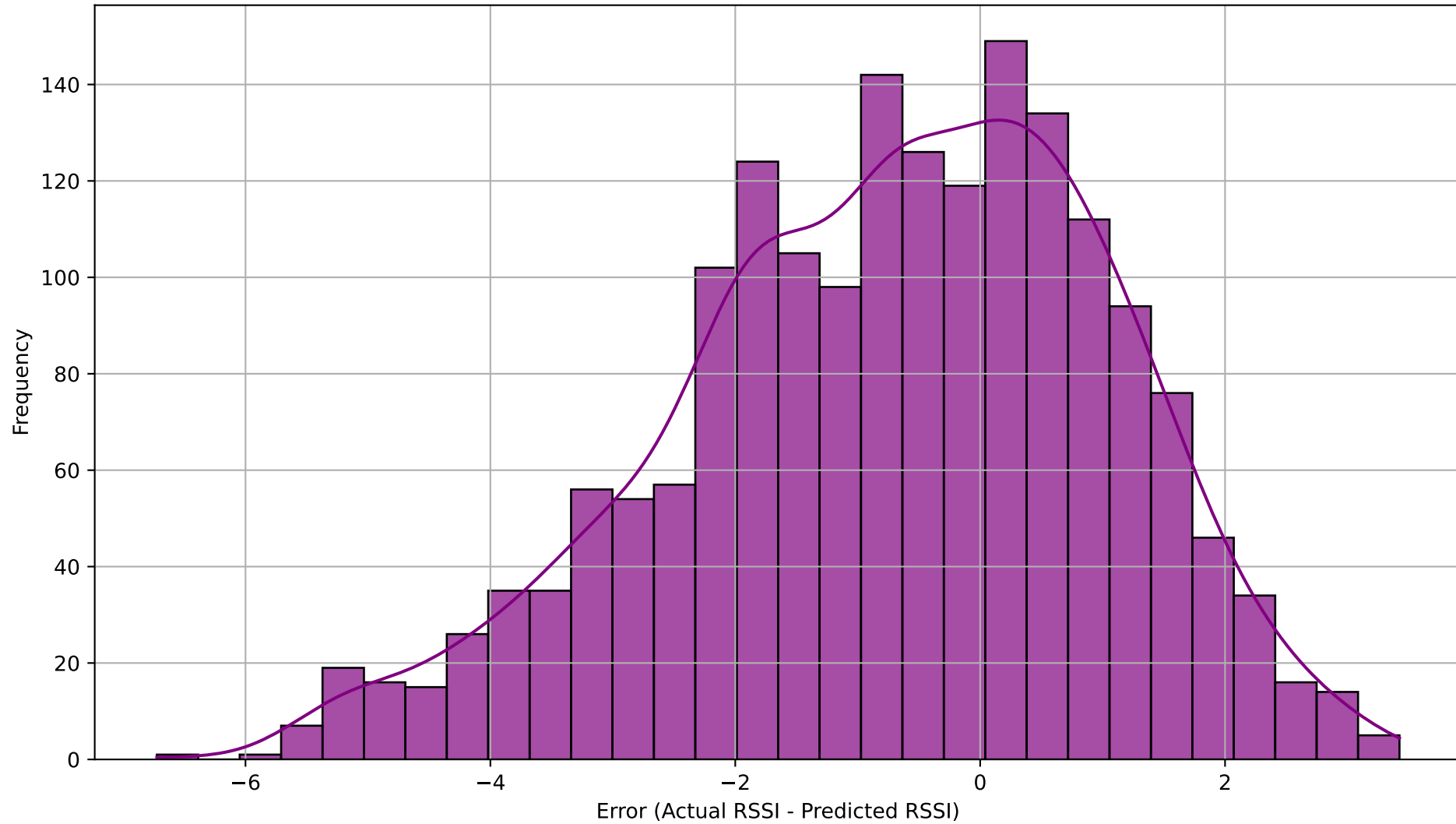
Actual vs Predicted RSSI (First Step)



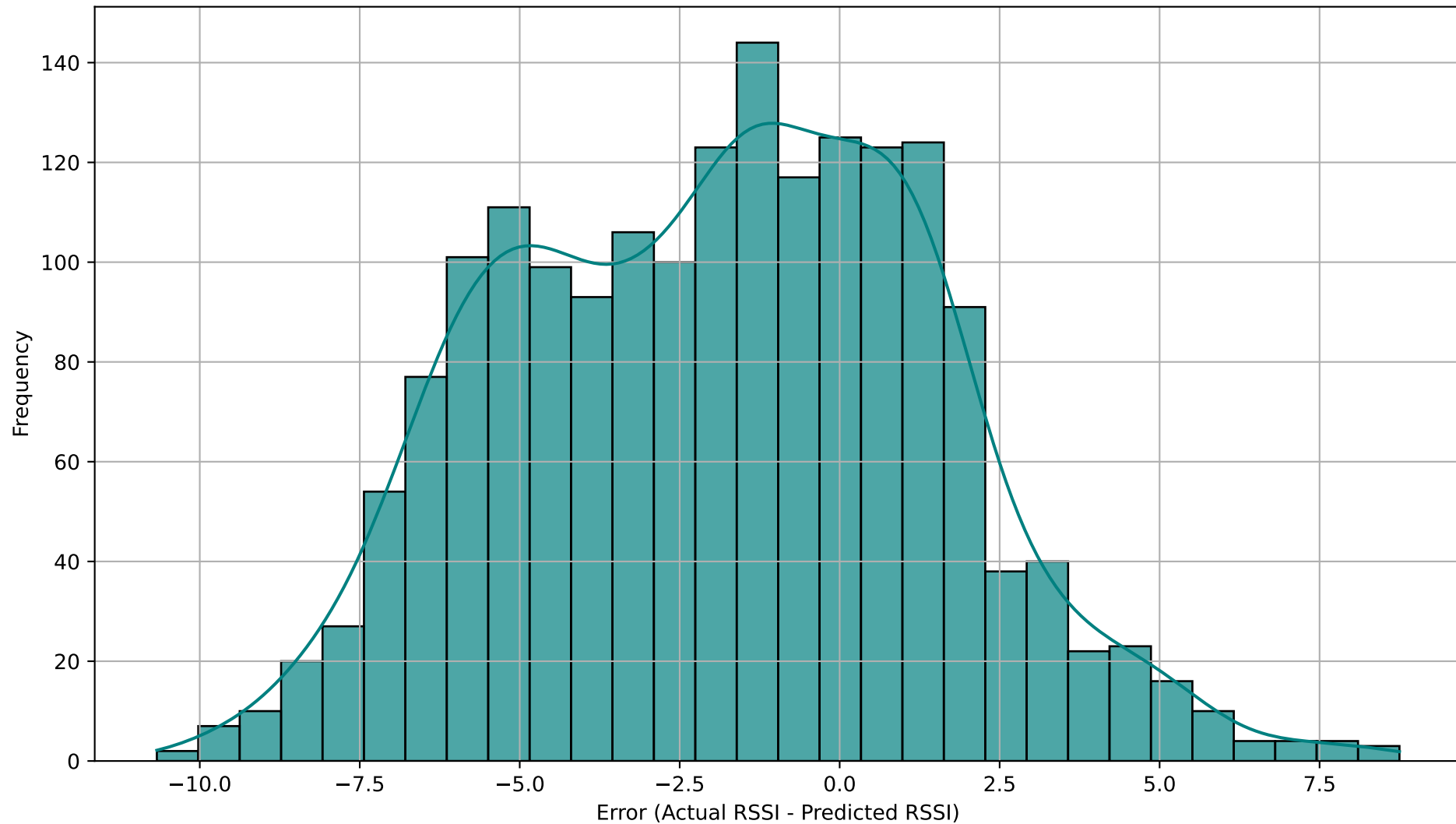
Actual vs Predicted RSSI (Last Step - step 48 $\approx$ 4.0 s)



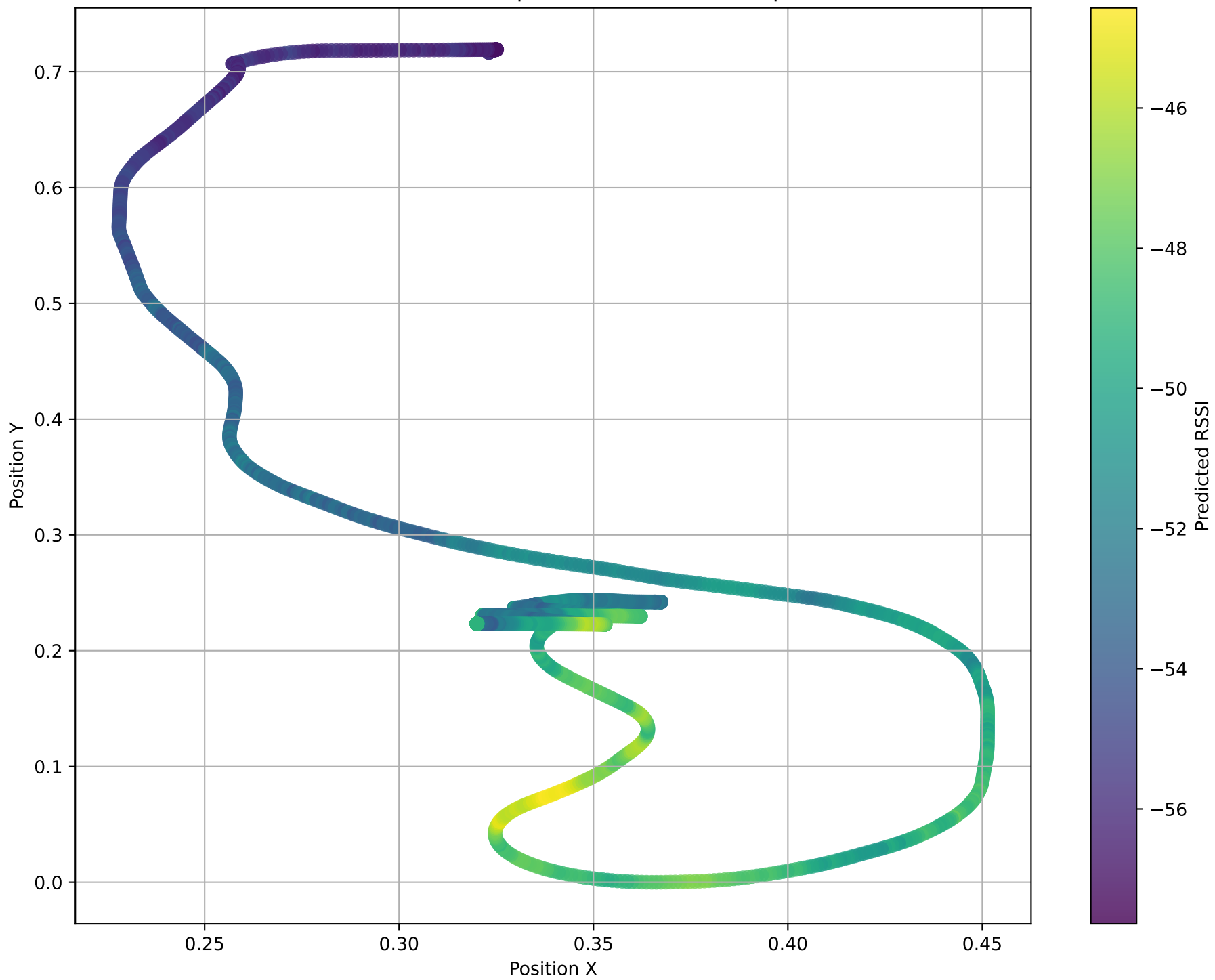
Distribution of Prediction Errors (First Step)



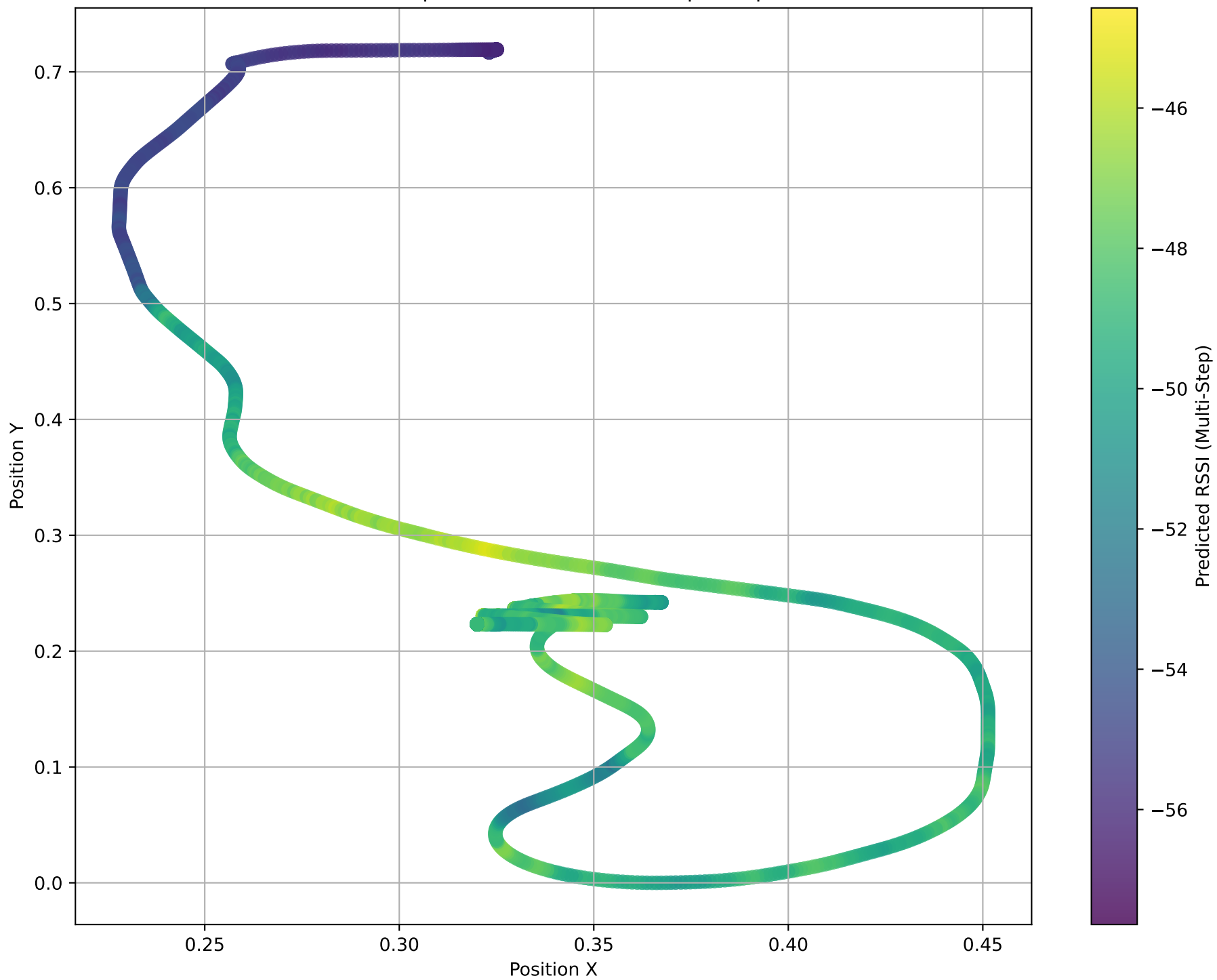
Distribution of Prediction Errors (Last Step - step 48 $\approx$ 4.0 s)



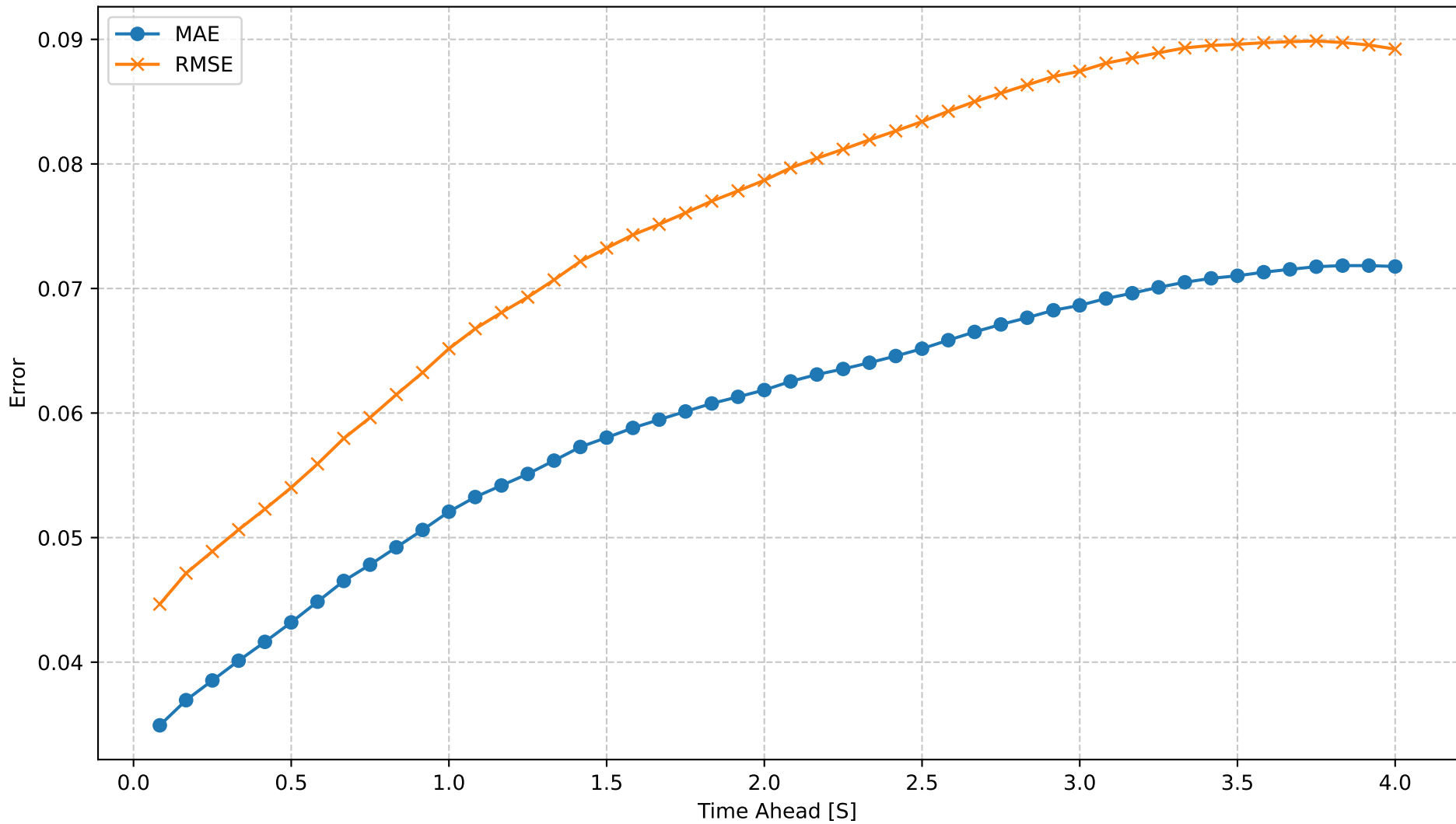
Predicted RSSI in Spatial Context (First Step)



Predicted RSSI in Spatial Context (Multi-Step - step 48 $\approx$ 4.0 s)



Multi-Step Forecast Error by Time Ahead



Resumo Automático

Dataset: rssi_odom_dataset_12apr25_ajust.parquet

Modelo: GRU

Forecast steps: 48

MAE / RMSE (dB)

- 1º passo : 0.03 / 0.04
- Mediana : 0.06 / 0.08
- 48º passo : 0.07 / 0.09

Crescimento do erro: 105.5% → Degrada Rápido (> 50 %)

3 piores passos (MAE): 46 (0.07), 47 (0.07), 48 (0.07)

Recomendação:

Considere aumentar o histórico de entrada ou usar um modelo mais profundo para melhorar multi-step.